



Elier Garcia

ELECTRICAL ENGINEER |
DATA & AI

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Spain

About me

My foundation is in electrical engineering. After a year managing electrical estimates and another supervising large-scale infrastructure for Microsoft and BMW Group, I gained an intuitive grasp of complex power systems and high-stakes operational friction.

Backed by a top-ranked Master's in Business Analytics and AI, I combine that practical field background with automated workflows and data-driven rigor to ensure continuous, resilient operations across critical infrastructure.

Skills

- **Operations & Systems:**
Electrical Infrastructure, Technical Drawings, Subcontractor Management, Root Cause Analysis
- **Automation & Scripting:**
Python, n8n (Workflow Automation), Docker, Git/GitHub
- **Data & Cloud:** SQL, PostgreSQL (Supabase), ETL & Data Pipelines, GCP (Firebase)
- **Applied Analytics & AI:**
Predictive Modeling, Machine Learning, Applied LLMs

Languages

Spanish: Native
English: B2

Certificates

Electric Power Systems
University at Buffalo (SUNY) ·
Coursera
Feb 2023

Work Experience

Full-Stack AI Developer

Freelance | Madrid | August 2025 — Present

- Engineered production-ready AI pipelines and RAG systems across multiple applications using Gemini, Firebase, and Supabase pgvector, implementing semantic chunking, dual-layer caching, and concurrency controls to optimize cost and latency.
- Designed asynchronous, event-driven automation workflows with n8n and webhooks, decoupling heavy inference and document processing tasks from client request-response cycles.
- Built full-stack web architectures (React, Next.js, Firebase) integrating external APIs and validation layers to eliminate hallucinations and deliver reliable, low-cost operational tools.

Electrical Construction Site Supervisor

DEMEK S.A de C.V. | Querétaro, Mexico | January 2024 — February 2025

- Promoted at six months to lead the full electrical scope of a Microsoft Data Center, overseeing crews of up to 60 personnel across medium voltage systems, substations, generator sets, and pump room installations.
- Designed from scratch a materials traceability system, combining SAP consumption exports with comparative Excel tracking, for the 19,847 m² BMW Group plant expansion, achieving full oversight across all 8 project sections.
- Maintained zero supply interruptions over 14 cumulative months of execution, managing technical RFIs for low and medium voltage systems and grounding installations across industrial facilities.
- Directed the installation, interconnection, and quality verification of critical power distribution equipment including diesel generators, modular UPS systems, dry-type and oil-filled outdoor transformers, MCCs, distribution boards, and PDUs, executing redundant A/B power cabling architectures to ensure uninterrupted operations.

Electrical Budget Analyst

DEMEK S.A de C.V. | Chihuahua, Mexico | January 2023 — January 2024

- Owned the full estimating cycle: AutoCAD take-offs → Excel quantity surveys → OPUS cost modeling → supplier negotiation → client presentation → change order management during execution.
- Delivered 30+ proposals (up to USD 500K) with an average cost deviation below 3%, consistently within 48-hour turnaround windows. Terex Project (Nuevo León): sole proposal to reach final presentation stage, delivered directly in English to the client. DEMEK was awarded and executed the full plant.

Education

Master's Degree in Business Analytics & Artificial Intelligence

Inesdi Business TechSchool | Madrid, Spain | April 2026

- Specialized in ETL/ELT pipelines, RAG architectures, applied Machine Learning, and Power BI data visualization.
- Graduated top of class with a GPA of 8.94/10 (Highest Academic Record Award).

Bachelor's Degree in Electrical Engineering

Instituto Tecnológico de Chihuahua | Chihuahua, Mexico | June 2023

- Major in Electrical Installations. Graduated with a GPA of 9.27/10.
- President of the Electrical Engineering Student Committee.